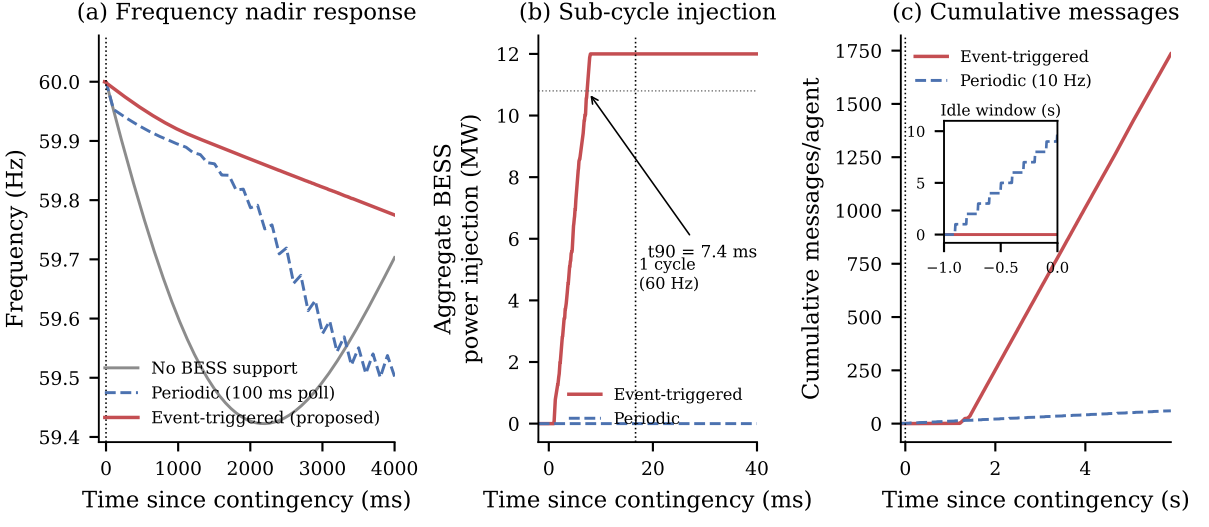
(a) Frequency nadir response
Frequency (Hz)
60.0
59.9
59.8
59.7
59.6
59.5
59.4
No BESS support
Periodic (100 ms poll)
Event-triggered (proposed)
0
1000
2000
3000
4000
Time since contingency (ms)
(b) Sub-cycle injection
Aggregate BESS power injection (MW)
12
10
8
6
4
2
0
t90 = 7.4 ms
1 cycle
(60 Hz)
Event-triggered
Periodic
0
20
40
Time since contingency (ms)
(c) Cumulative messages
Cumulative messages/agent
1750
1500
1250
1000
750
500
250
0
Event-triggered
Periodic (10 Hz)
Idle window (s)
10
5
0
−1.0
−0.5
0.0
0
2
4
Time since contingency (s)